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Differences in Usability Between Three Date-entry Designs

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Abstract

In this study, we explored the extent to which the design of date-entry fields (e.g., text entry, dropdown, and calendar) interacts with characteristics of the task (e.g., knowing the day of the week, relying on memory) to affect the usability (effectiveness, efficiency, and satisfaction) with which participants complete date-entry tasks for near dates. A total of 1000 participants completed different tasks using one of the three date-entry designs. For some of the tasks, knowing the day of the week was required to provide a correct response. Our findings indicate an advantage of the calendar design over the text entry design in terms of effectiveness and over both text entry and dropdown designs for most tasks in terms of efficiency. No difference across designs was observed for perceived difficulty, with most participants rating all date-entry design conditions as “extremely easy”. Based on our findings, we recommend implementing a calendar date-entry design for near dates, particularly for tasks requiring the day of the week. Future studies can expand these findings by exploring whether the advantage of the calendar design over the text entry and dropdown designs holds for distant dates.

Keywords: Date-picker, date-entry, usability, drop-down, text-entry, calendar.

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EXECUTIVE SUMMARY

This paper examines how three common date-entry designs—text entry, dropdown menus, and calendar interfaces—perform when users enter dates that are close to the current date. Using two separate experiments with a combined sample of 1,000 U.S. residents, the authors assess each design’s effectiveness (accuracy), efficiency (time-on-task), and satisfaction.

Across both studies, the calendar design consistently demonstrated the strongest overall usability for near-date tasks. It produced the highest average accuracy, though statistically significant differences were limited to specific tasks. More notably, the calendar design delivered the fastest performance on nearly all tasks, outperforming dropdowns and text entry even when tasks differed in structure—whether framed as an actual date, a day of the week, or a date recalled from memory.

Satisfaction ratings showed that participants generally found all designs easy to use, with no statistically significant differences across conditions. Although the calendar showed a slight trend toward higher ease-of-use ratings, the study’s satisfaction results were not robust enough to establish clear superiority.

Overall, the findings indicate that for online forms requiring respondents to enter near dates, the calendar design offers meaningful advantages—particularly in efficiency. The results also highlight how specific task characteristics influence usability outcomes and reinforce that date-entry design cannot be evaluated independently from the nature of the task itself.

1. INTRODUCTION

Online forms collect factual data on a wide range of topics (e.g., a person’s age, last time they visited a doctor, etc.) and the answer to some of these questions require respondents to provide a date (Bargas-Avila et al., 2011; Persson, 2020; Wang, 2022). Some of the most common date-entry designs include text entry boxes, dropdown options, or calendars. The characteristics of these designs may interact with the characteristics of the date to be entered. For example, whether the date to be entered is close to the date of the study (a near date) or farther away from it (a distant date).

One aspect that has been overlooked in the literature is the fact that some date-entry tasks may be easier to complete when respondents know the day of the week. For example, a patient trying to select a doctor’s appointment may prefer to select “next Wednesday” instead of “the 15th of January.” In this study, we explored the extent to which some date-entry designs (e.g., text entry, dropdown, and calendar) affect the usability, including the effectiveness, efficiency, and satisfaction with which respondents enter near dates. We vary this based on questions that may or may not require them to know the day of the week (e.g., entering the date for “last Friday” or entering the date “April 21, 2022”) or may require them to rely on their memory (e.g., “your last doctor’s appointment”).

2. BACKGROUND

2.1 Date-entry Designs

In this section, we present a description of some of the most common date-entry designs (text entry, dropdown, and calendar) along with a short summary of the research that has explored their effectiveness and efficiency. Because different authors may use the same label for different date-entry designs (e.g., date picker for calendar and spinner for a dropdown on mobile

devices; Antoun et al., 2017; Persson, 2020; U.S. Web Design System, 2025), this section provides a label, a description, and an image to help the reader visualize the design being described.

2.1.1 Text entry Design

The text entry design allows respondents to type the desired date, typically into a fixed-length field where an underline is presented with a pre-specified number of spaces for character entry. One potential issue with this design is that respondents may not enter dates in the desired format (Bargas-Avila et al., 2011; Wang, 2022). For example, users may either switch the order of the numbers for day and month (e.g., entering 01/06 instead of 06/01 to report June 1st), or may not enter the requested number of digits (e.g., entering “1” as opposed to “01” for January or entering “24” instead of “2024”). To solve these potential problems, designers may use a mask format, where the response field contains format cues about the information to be entered. Figure 1 provides a visualization of a text entry design with a fixed-length field without masking (left) and with masking (right).

Figure 1. The text entry design may not specify the format (left) or may include a mask to specify the format for date-entry (right)

Date format not-specified	Date format specified
___ / ___ / ____	mm / dd / yyyy

2.1.2 Dropdown Design

This design consists of three fields with pre-specified values for month, day, and year. The “month” field can be the month names, the month number, or a combination (e.g., 1-January). The “day” field provides the numbers 1-31 (depending on the selected month), and the year field provides a range of years, typically in four-digit format (except for credit card

expiration dates, which typically require only the last two digits). This design allows the user to select each value by scrolling through the list and then selecting the applicable choice. For mobile device operating systems, the equivalent of a dropdown design may be called a “spinner,” “picker,” or “spin wheel” (Antoun et al., 2017). In this paper, we refer to all these designs as “dropdown.”

The advantage of the dropdown over the text entry design is that it prevents users from entering the data in the wrong format or in the wrong order. One potential disadvantage of this design is the amount of scrolling needed for a year in the distant past, such as the year of birth for an older individual. Another disadvantage is that, as can be observed in Figure 2, an expanded dropdown box may conceal other information in the screen.

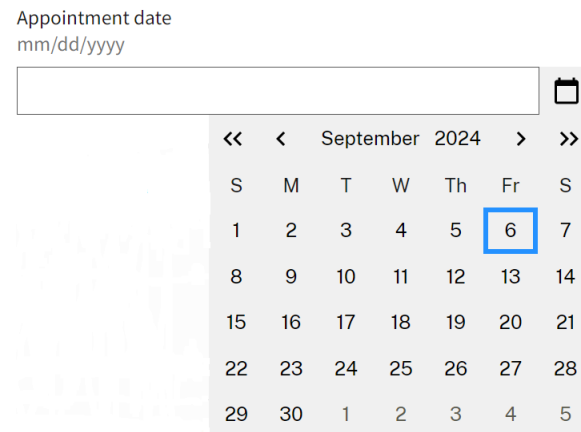
Figure 2. Dropdown design with the month field collapsed (left) and expanded (right).

Figure 2 displays two versions of a date selection form. The left version shows the 'Month' dropdown menu collapsed, displaying only the text 'Month'. The right version shows the 'Month' dropdown menu expanded, displaying a list of months from January to December. Both versions include a title 'Select last Friday's date.' and three input fields for 'Day' and 'Year'.

2.1.3 Calendar Design

The calendar design includes a pop-up display of the current year and month, and a highlighted current day. Common calendar designs include buttons or arrows that allow respondents to change the month or the year, as shown in Figure 3.

Figure 3. The calendar design with arrows for navigation to different months or years



Note: Image taken from U.S. Web Design System:
<https://designsystem.digital.gov/components/date-picker/>. Accessed 01/31/2025

Although the calendar design is very efficient when the date to be selected is in the default month and year (only one click necessary), this design becomes cumbersome if the year to be selected is not the current year shown in the calendar. While the example in Figure 4 includes separate arrows to change the year and month, other calendar designs may only include arrows to navigate one month at a time, resulting in the respondent having to click several times to change the year.

3. LITERATURE REVIEW

Studies that have explored the usability of different design options for entering dates have found conflicting results for both effectiveness and efficiency outcomes.

Regarding effectiveness, although some studies have found no difference in number of errors between the text entry and calendar designs (Persson, 2020) or text entry and dropdown designs (Waloszek et al., 2020), others have found that text entry leads to a higher number of errors than both calendar and dropdown (Myka et al., 2019). However, other studies have found

that the higher number of errors observed in the text entry design is limited to errors involving the participants switching the numbers for day and month (Bargas-Avila et al., 2011), or when the date to be entered is a near date (a day in the same month; Drygielski et al., 2018).

When focusing on differences in effectiveness between the calendar and dropdown designs, some studies have found that these two designs lead to similar number of errors (Myka et al., 2019; Türkcan & Durdu, 2018), but other studies have found that the calendar leads to a higher number of errors than the dropdown design (Bargas-Avila et al., 2011). Yet, other studies have found that the calendar design not only outperforms the dropdown design, but it also leads to no errors (Waloszek et al., 2020), although this advantage of the calendar design may be limited to tasks involving near dates (Drygielski et al., 2018). Furthermore, Türkcan and Durdu (2018) found that the number of errors committed by participants using the calendar design for distant dates was so high that this was the only condition where some participants did not complete the task.

Regarding efficiency, although some researchers have not observed a difference in time-on-task between text entry and calendar (Persson, 2020) or text entry and dropdown (Drygielski et al., 2018), other researchers have observed that text entry outperforms both the dropdown and calendar designs (Bargas-Avila et al., 2011; Myka et al., 2019; Türkcan & Durdu, 2018; Waloszek et al., 2020). However, the higher efficiency of the text entry design may be limited to distant dates (Drygielski et al., 2018).

When comparing the dropdown and calendar designs in terms of efficiency, some studies have observed no difference in time-on-task between these two designs (Bargas-Avila et al., 2011; Myka et al., 2019). Others have observed a shorter time-on-task for the calendar than for the dropdown design (Waloszek et al., 2020), yet others have observed the opposite pattern

(Türkcan & Durdu, 2018). However, studies that have manipulated the date to be entered (near or distant) have observed an interaction between date-entry design and the date: When the date to be entered is near, the calendar design performs better than the dropdown design in terms of efficiency. However, when the date to be entered is distant, the pattern is reversed, and the dropdown design results in lower time-on-task than the calendar design (Drygielski et al., 2018).

Some of the conflicting findings in the literature may be explained by differences in methodology in terms of sample size, and manipulation of different factors. For example, the study conducted by Myka et al. (2019) consisted of 10 participants per design, and no inferential analyses were conducted. Therefore, it is not clear whether the differences in time and errors between the designs are due to random variability or due to specific characteristics of the designs.

The lack of difference in number of errors between text entry and calendar design observed in the Persson (2020) study is contrary to the findings of Myka et al. (2019) and Türkcan and Durdu (2018). The conflict between these studies may be explained by the fact that Türkcan & Durdu, included both near and distant dates in the task, while the task in the Persson (2020) study did not require participants to change the year. In fact, Persson (2020) acknowledged that the results may have been different if the task would have required participants to change the year.

Finally, the conflicting results between Drygielski et al. (2018) and other studies (Myka et al., 2019; Persson, 2020; ; Türkcan & Durdu, 2018; Waloszek et al., 2020) is due to differences in operationalization of outcomes. The Drygielski study is the only study where the operationalization of errors takes into consideration whether the numbers for day and month were switched, while other studies did not make this distinction.

Regardless of these limitations, the studies described above have expanded our knowledge by allowing the conclusion that there is not a “one size fits all” when it comes to date-entry designs. The current literature drives attention to the importance of not only considering the design for date-entry, but also considering the characteristics of the stimuli (e.g., near and distant dates: Drygielski et al., 2018) and the operationalization of outcomes (e.g., different types of errors: Bargas-Avila et al., 2011).

One characteristic of the stimuli that has not been explored in the literature is the day of the week. Although the day of the week may not be necessary for some type of dates (e.g., date of birth), it may be indispensable for other situations. For example, a patient may ask the staff at the doctor’s office to reschedule their appointment for “next Wednesday”.

For tasks when knowing the day of the week may be advantageous, the calendar design may be the most effective and efficient given that the day of the week is visible to the respondent; that visibility is a design characteristic not included in any other design. To the best of our knowledge, no study has compared the text box or dropdown menu designs to the calendar design in situations where knowing the day of the week would be helpful to select the correct date.

The Present Study

The purpose of the present study is to add to the existing literature by exploring the usability associated with three common date-entry designs (text entry, dropdown, and calendar), focusing on entering near dates. The study investigates whether the usability of the design (effectiveness, efficiency, and satisfaction) depends on whether the day of the week is part of the date-entry task (e.g., entering last Friday), when the date relies on memory (e.g., entering your

last doctor's appointment), or a date either in the immediate future or close past. For this purpose, we developed the following research questions:

1. Is there a difference in effectiveness as measured by data entry accuracy between text entry, dropdown, and calendar designs when entering near dates, when the day of the week is provided (e.g. last Friday), or when the actual date is provided (e.g. April 21, 2022)?
2. Is there a difference in efficiency as measured by time-on-task between text entry, dropdown, and calendar designs when entering near dates, when the day of the week is provided, when the near date is from memory, or when the actual date is provided?
3. Is there a difference in satisfaction for the three designs for near dates?

To answer these research questions, we conducted two studies that required participants to enter near dates using either a calendar design, a text entry field, or a dropdown menu to complete tasks that may or may not have required them to know the day of the week.

4. METHODOLOGY

4.1 Sample

The data for both studies were collected in 2022. The Office of Management and Budget reviewed and approved the materials and methodology for this study. The first study was conducted in April and May and the second study was conducted in July. Two separate quota samples of 500 U.S. residents from a nonprobability panel completed each study (for a total of 1000 participants). All participants received informed consent prior to the start of the study. The first study oversampled bilingual participants (English-Spanish; for a separate research objective) which resulted in more Hispanic participants. The second study required participants to use a

personal computer to complete the tasks (again for a separate research objective). Table 1 includes basic participant demographics and device used for each study.

Table 1. Participant characteristics and device used. Each study n = 500.

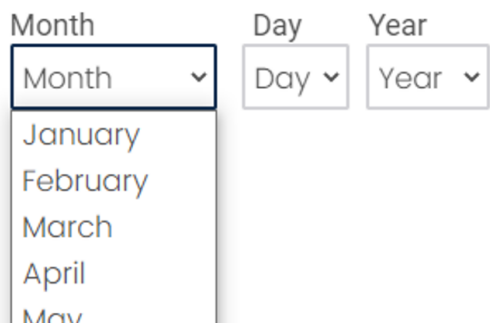
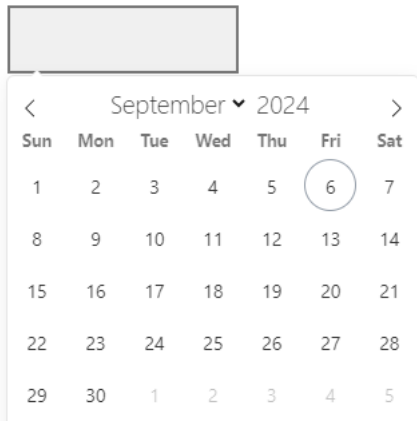
Variable	Study 1	Study 2
Age		
Mean	43	47
Range	18 to 87	18 to 98
Sex		
Male	39%	45%
Female	61%	55%
Race		
White	66%	78%
Non-White	34%	22%
Ethnicity		
Hispanic origin	49%	7%
Not Hispanic	51%	93%
Device used		
Desktop/laptop	70%	100%
Mobile	30%	--

4.2 Experimental Design

Both studies used a between-subjects design with the factor “Date-entry design” including the three conditions: text entry, dropdown, and calendar. Participants were randomly assigned to each of the experimental conditions. These conditions are described in Table 2.

Table 2. Description and visualization of the three levels for the “Date-entry design” factor

Design	Description	Visualization of design
Text entry	Text field with a label (MM/DD/YYYY) above the field and a mask in the field that would not disappear as the user typed.	

Dropdown	Three separate fields, Month, Day, and Year with labels above each field. The months were spelled out fully and the years were ordered from 2022 back.	
Calendar	When the user entered the field, the current month appeared with the current day circled. Participants could navigate the months by the dropdown on the month label and the years by the arrows.	

4.3 Tasks and Procedure

The experimental tasks were part of an 8-minute online questionnaire that required participants, among other tasks, to answer questions about demographic characteristics. Each of the experimental tasks were presented in a separate screen. Tasks were programmed and presented to participants using the Qualtrics survey platform.

In Study 1, participants performed the following date-entry tasks in the order below:

- 1) Pretend you want to set an appointment for one week from today. Go ahead and enter that date using the design below.
- 2) Now enter April 21, 2022.
- 3) Now enter May 21, 2022.

After completing the experimental tasks, participants answered an ease-of-use question, “How easy or difficult was it for you to enter these three dates?” using a 5-point labeled scale from 1 - *Extremely easy* to 5 - *Extremely difficult*.

Study 2 included new tasks, two of which included a day of the week in the instruction and one task which relied on the participant’s memory. Again, the tasks were in the same order for every participant and on their own screen. Due to experimenter’s error, data was not collected for the ease-of-use question in Study 2.

- 4) Enter the date of your last doctor’s appointment.
- 5) Enter last Friday’s date.
- 6) Enter the date of two Mondays ago.

Assuming that the last doctor’s appointment was within the year, all tasks focused on near dates for both studies. Some dates were in the same month as the study completion, some were in the next month, and some were in the previous month.

4.4 Outcomes of Interest and Analysis

4.4.1 Effectiveness

Effectiveness for this study was defined as accuracy and it was recorded as a binary variable (accurate or not accurate). For the *one week from today* task in Study 1, accuracy was based on whether the date entered was one week from the date the participant completed the study, which was data recorded by Qualtrics. We calculated average accuracy by dividing the number of correct responses by the total number of participants that answered the question and conducted a t-test to determine significance. The same type of calculation was used when determining accuracy for entering *last Friday’s date* and entering *two Monday’s ago* in Study 2. For the other two tasks in Study 1, the date entered was accurate if it was *April 21, 2022*, for the

second task and *May 21, 2022*, for the third task. Task 4 (entering the date of the last doctor's appointment) is not included in this analysis since we do not know the true answer.

Analysis for each study was conducted separately but used the same mixed-logistic model with a random effect for the participant, each of which responded to multiple date-entry tasks:

$$\pi_i = \beta_0 + \beta_1[\text{condition}]_i + \beta_2[\text{demographics}]_i + \beta_3[\text{question}]_i + \beta_4[\text{condition} * \text{question}]_i + e_i,$$

where π is the success probability of an accurate response, i is the respondent level indicator, and

$$\beta_0 = \gamma_{00} + \mu_0,.$$

The three conditions are text entry, dropdown, and calendar. The demographic variables are the participant's age, sex, Hispanic origin, race, and, for study 1 participants only, the device used.

4.4.2 Efficiency

Efficiency was measured by time-on-task (seconds) for each correctly completed task. Time was measured from the time the page loaded until the participant advanced to the next page. We used a mixed-linear model with the log of time as the dependent variable, the condition as the independent variable, and the same covariates as were used in the accuracy model. Again, a random effect for the participant was added to account for the number of date-entry tasks.

For both effectiveness and efficiency, we report significant pairwise comparisons after using a Bonferroni correction where the F test of the interaction between the condition and the task was significant.

4.4.3 Satisfaction

Satisfaction was measured by the ease-of-use question in Study 1. The three lowest scores were combined into one response choice since very few participants reported that the

tasks were difficult, then a Chi-Square statistic was calculated to determine if the ease of use was independent of condition.

4.5 Hypotheses

Hypothesis 1 - Effectiveness: The date-entry design will be a significant predictor of accuracy. Answers using the calendar will be more accurate for the tasks indicating the day of the week (tasks 1, 5, 6) compared to the text entry or dropdown. Answers for the tasks with a date only (tasks 2 and 3) will be more accurate in the calendar and dropdown condition than the text entry as suggested by earlier studies.

Hypothesis 2 - Efficiency: The date-entry design will be a significant predictor of time-on-task. As observed by Drygielski et al. (2018), and given that the current study involves near dates, the calendar will be faster to use than dropdown and text entry for all tasks.

Hypothesis 3 - Usability: Given support for hypotheses 2 and 3, the calendar design will be reported as easier to use than the other two designs.

5. ASSUMPTIONS AND LIMITATIONS

We assume that participants' report of their last doctor's appointment was reported accurately and did not verify this information. It is possible that participants may have just estimated this date.

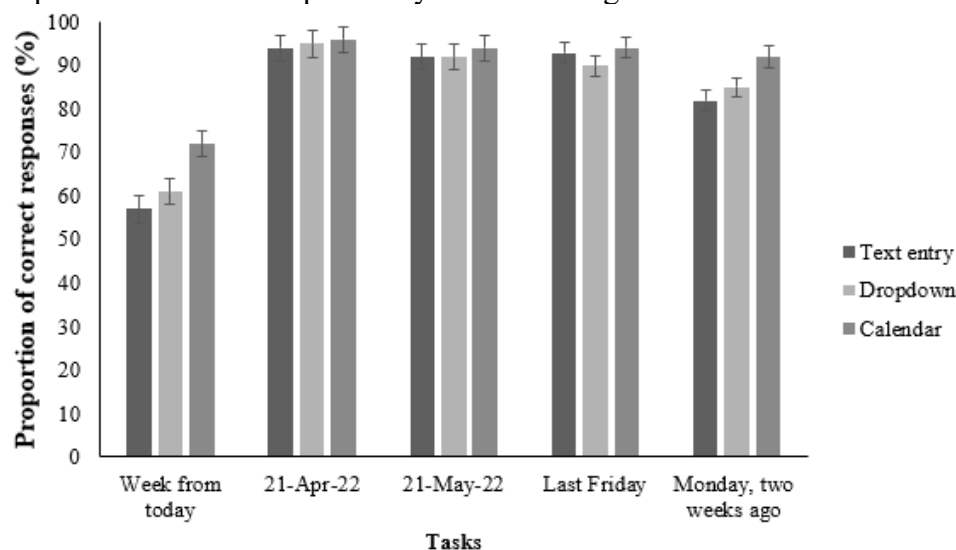
6. RESULTS

6.1 Effectiveness

The calendar design had the highest average accuracy across all five tasks. However, it was only significantly higher for a few of the tasks. The calendar resulted in more accurate data entry than the text entry design for the *one week from today* task ($p < .01$) and the *two Mondays ago* task ($p = .06$). There was no significant difference in accuracy between the calendar and the

dropdown for any of the tasks. Similarly, there was no significant difference in accuracy between text entry and dropdown for any of the tasks. Figure 4 provides a visualization of the effectiveness data.

Figure 4. Proportion of correct responses by task and design condition



Note. 500 participants completed each question (they were divided equally among the conditions). Error bars represent the standard error of the mean.

6.2 Efficiency

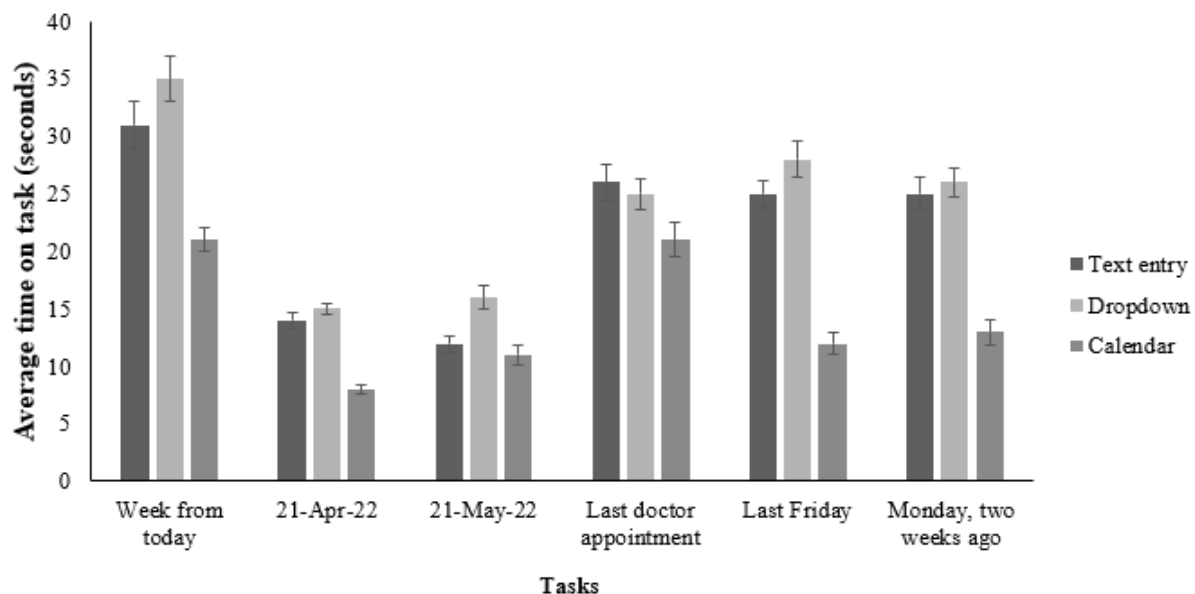
The calendar design was faster to use than the dropdown design across all tasks.

Participants entered data more quickly with the calendar than the dropdown for the *one week from today* task ($p < .01$); the *April 21* task ($p < .01$); the *May 21* task ($p < .01$); the *last doctor appointment* task ($p < .01$); the *last Friday* task ($p < .01$); and the *two Monday's ago* task ($p < .01$).

The calendar design was also significantly faster to use than the text entry design for 5 of the 6 tasks. Participants entered data more quickly using the calendar than the text entry design for the *one week from today* task ($p < .01$) and the *April 21* task ($p < .01$); the *last doctor appointment* task ($p < .01$); the *last Friday* task ($p < .01$); and the *two Monday's ago* task ($p < .01$).

Text entry was faster to use than dropdown for only the third task in Study 1: the *May 21* task ($p < .01$). None of the time comparisons between dropdown and text entry for any of the other questions were statistically significant. As shown in Figure 5, on average, the calendar appears to take the least amount of time across all tasks, followed by text entry, and then the dropdown design.

Figure 5. Time on task by task and design condition

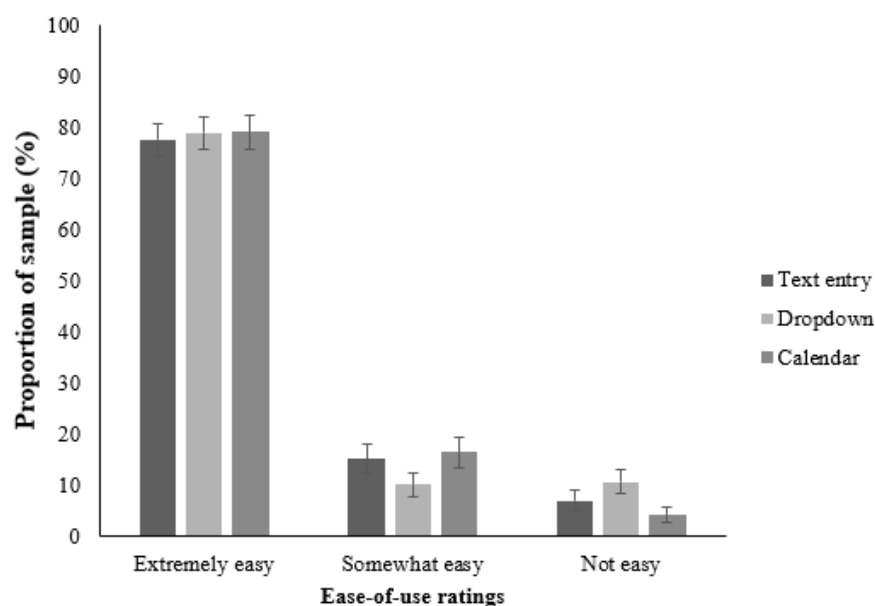


Note. Study 1: correct trials=1,300; Study 2: correct trials=1,400. Correct trials rounded to the nearest 100. Error bars represent standard error of the mean.

6.3 Satisfaction

Although there was a trend for calendar design to show higher ease of use scores, the Chi-square test of independence was not significant at the .05 level across conditions in Study 1. Figure 6 provides a visualization of the ease-of-use data.

Figure 6. Proportion of sample that selected the different ease-of-use ratings for each text entry design condition.



Note. $n = 500$. Error bars represent the standard error of the mean.

7. DISCUSSION AND CONCLUSIONS

Many online forms require respondents to enter dates. Some dates, such as birthdates, can be a long time ago while others, such as reservations or appointments, can be close to the current date. There are multiple designs available to collect this information, each with their benefits and drawbacks. This analysis aimed to determine which design has the highest accuracy and efficiency in situations where the date is near the current date. The results from both experiments indicate that the calendar design is overall the best option for entering near dates between the three designs tested. These results are consistent with prior findings from Waloszek et al. (2020) and Drygielski et al. (2018).

The first hypothesis was that the calendar would result in more accurate entry for tasks indicating the day of the week compared to the text entry and dropdown, and that text entry

would result in lower accuracy compared to the other two designs when a date was given. While there was a trend suggesting the calendar design results in more accurate data for near date tasks that involve day of the week and dates, there were few differences that were statistically significant, and this hypothesis was only partially supported.

Participants completed three tasks referring to a day of the week (rather than a date). The calendar resulted in more accurate entry for two of those three tasks compared to text entry. There was no statistical evidence that the calendar resulted in more accurate entry compared to dropdown. There was also no statistical evidence that accuracy diminished with text entry compared to the calendar or the dropdown when a date was provided in the task. This finding matches the Persson (2020) finding but is contrary to Myka et al. (2019), Waloszek et al. (2020) and Drygielski et al. (2018). Those three papers found that text entry results in more inaccurate entry. In the present experiment, the mask used in the text entry fields and the label above the fields likely helped participants understand the desired format. It could be that these features supported participants so that they entered data as correctly as those who received the dropdown or the calendar design when the task included the date.

The second hypothesis was that the date-entry design will be a significant predictor of time-on-task and that the calendar will be faster to use than dropdown and text entry for near dates, similar to the findings by Drygielski et al. (2018). In all six tasks, participants used the calendar more quickly than the dropdown design, supporting the second hypothesis. Furthermore, in five out of six tasks, participants were able to answer with the calendar more quickly compared to text entry. This finding was consistent no matter if the task was presented as a day of the week, an actual date, or a date from memory.

Repetition could have been the reason the calendar and text entry timing were not different for one task. In Study 1, participants completed two similar tasks, one after the other: “enter April 21” and then, on the next page “enter May 21.” The calendar was quicker to use for the April 21 task than text entry, but not for the May 21 task. We suspect that the participants who received the text entry design became more efficient, eliminating any difference in task completion time between the text entry design and the calendar. In fact, they sped up so much in the second task that they were significantly faster than the participants who received the dropdown design.

The final hypothesis was that the calendar would be reported as easier to use than the other two designs. While the satisfaction data suggests the calendar might be easier to use, the results were not statistically significant. Thus, our hypothesis was not supported. Our satisfaction measure did have some limitations; it was only asked in one of the two studies, and had satisfaction been recorded after each task, that might have been more useful to tease out the difficulty of the task compared to the difficulty of the design.

Previous literature showed that calendars performed better than text entry and dropdown for date-entry on online forms, particularly for near dates. These experiments support those findings, with participants entering data into the calendar significantly faster than the other two designs and establishing a trend of having higher accuracy. For near date entries involving day of the week, we find some evidence that the calendar improves accuracy compared to the other two designs, but it is not as strong as the efficiency finding. Interestingly, the text entry condition performed better in the present experiments compared to what had been reported in the literature, both in time-on-task and accuracy. This is likely because a label was above the field and masking was used in the text field, which alerted the respondent to the desired format, thereby reducing

errors. This finding showcases how the evolution of designs can improve performance. Future research can explore whether small changes to existing designs, such as including a calendar icon next to the text field, can lead to additional gains in performance (U.S. Web Design System, 2025), or whether the advantage of the calendar design over the text entry and dropdown designs holds for distant dates.

8. REFERENCES

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